A Machine Learning Model for the Prediction of Hessian Matrices based on Semiempirical Electronic Structure Theory

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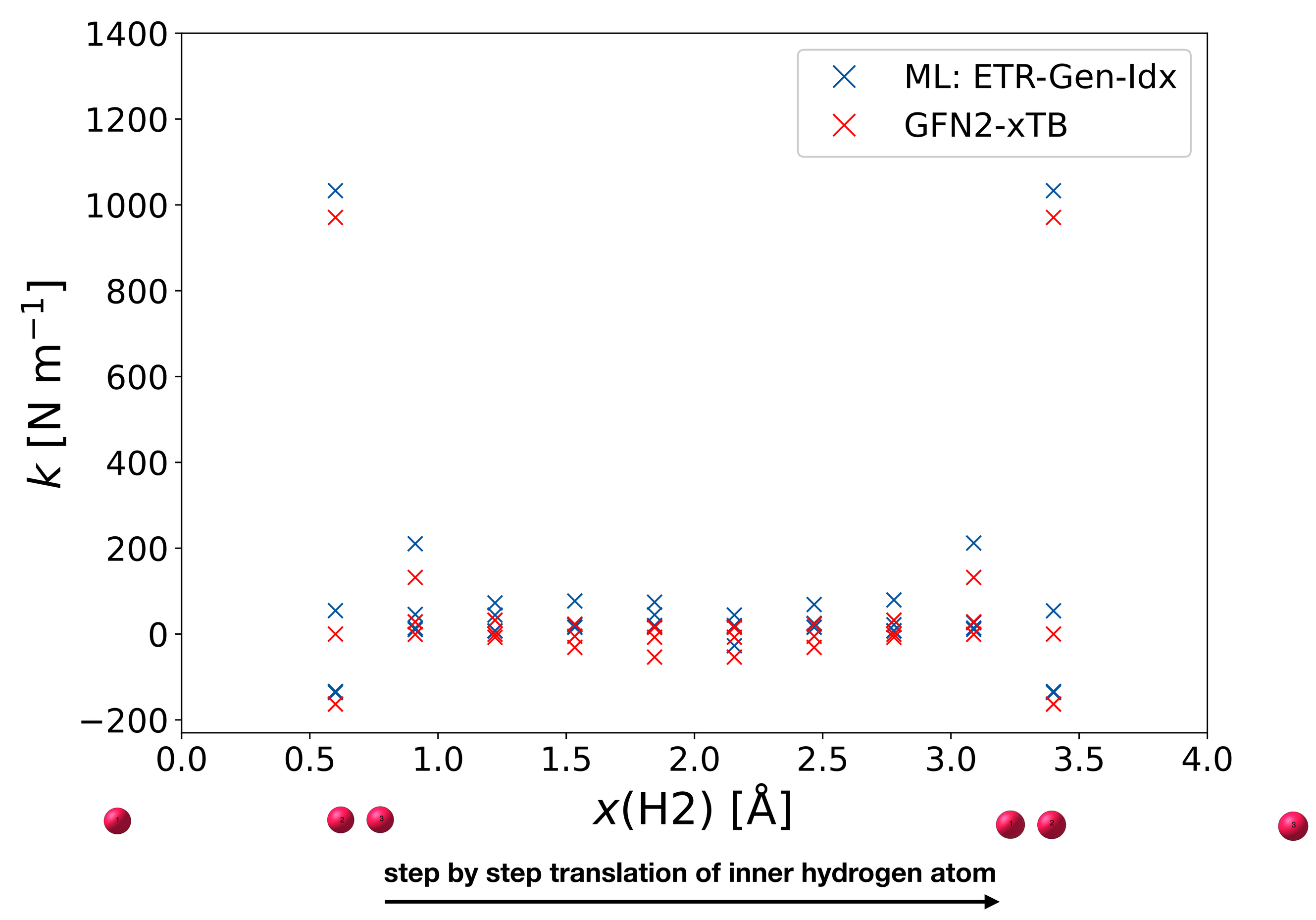
Introduction

- The Hessian matrix is a target of interest, due to its high computational cost and its versatile applications.
- Atomistic Features, bassed on GFN2–xTB quantities, are used for the prediction of the target Hessian matrix elements
- The features are permutation and rotational invariant to achieve a unique mapping onto the target.
- The zero point vibrational energy is used to quantify the precision of the model.

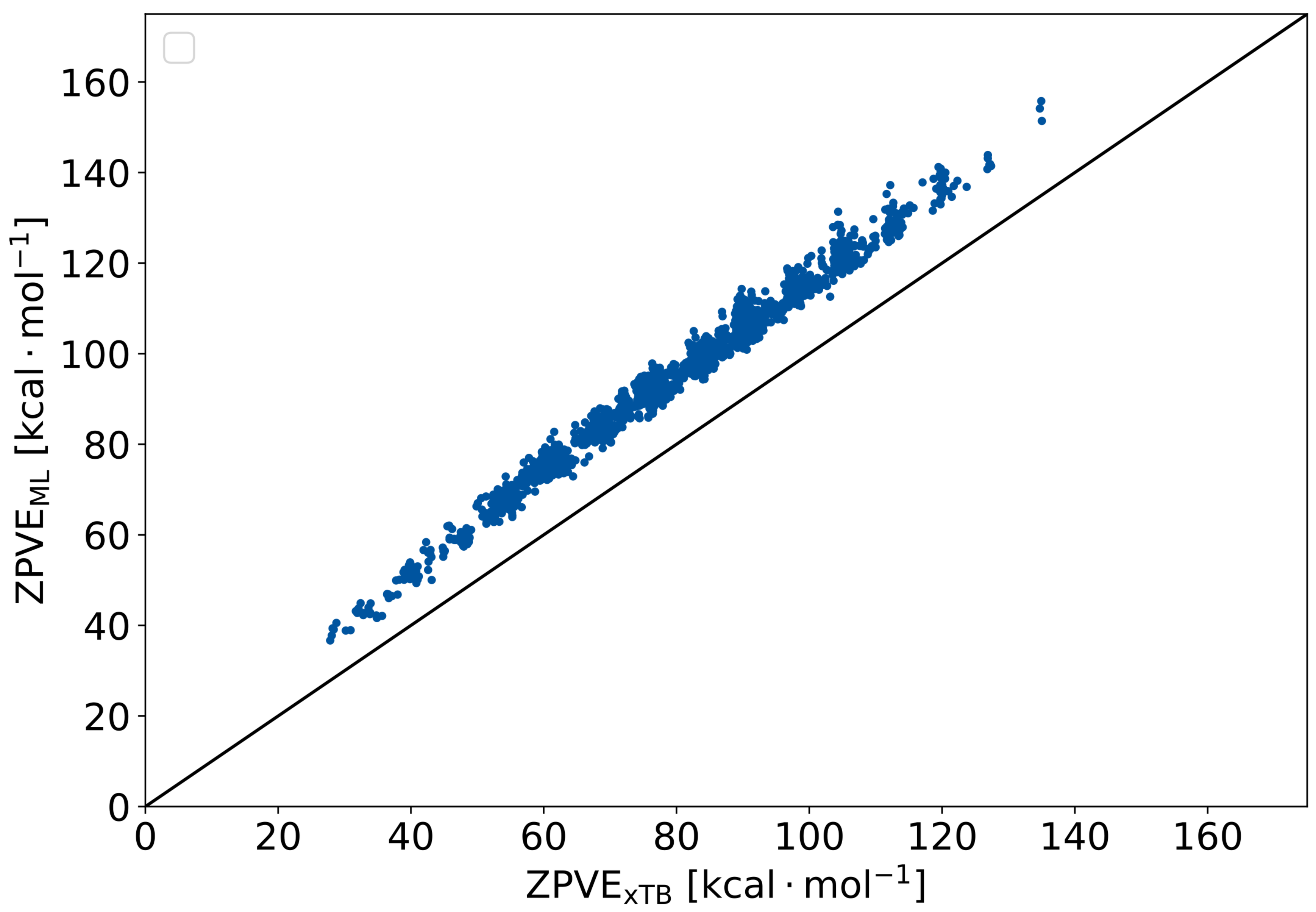
Feature Generation

- A
- B
- C
- D
- E
- F
- G

Verification of Unique Mapping



Predicted ZPVE of GBD7 Set



Conclusion and Future Perspective

C

References

[1] A
[2] B
[3] C
[4] D

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